

User Manual: American Sign Language (ASL) Vision Studio

1. Introduction

The ASL Vision Studio is an end-to-end computer vision pipeline designed to capture, annotate, train, and deploy real-time American Sign Language recognition models using YOLOv8 and MediaPipe.

2. System Requirements

- OS: Windows 10/11 (with camera hardware)
- Python 3.11
- Hardware: Dedicated GPU recommended for faster training
- OpenCV, Ultralytics YOLOv8, MediaPipe, PyYAML, Streamlit

3. Setup & Installation

1. Clone the project repository.
2. Install required libraries:

```
pip install ultralytics opencv-python mediapipe streamlit pyyaml
```

3.1 Repository Guidelines & Data Management

To maintain a lightweight repository, large binary files (e.g., trained '.pt' weight files) and raw training image datasets are intentionally excluded from version control. These files are specific to local training environments and are too large for efficient repository synchronization. Users should generate their own custom model weights locally using the provided training scripts.

4. Dataset Management

4.1 Adding New Alphabets

Update the 'dataset.yaml' file to include your new class labels under the 'names' dictionary. Each entry must have a unique integer ID.

4.2 Capturing Data

Run the capture tool. It dynamically reads from dataset.yaml and saves images in 'data/raw/[CLASS]/'.

```
python src/dataset_capture.py
```

4.3 Automated Annotation

Annotate the collected images using MediaPipe landmarks for YOLO format.

```
python src/auto_annotate.py
```

5. Training & Tuning

To train from scratch, use 'train.py'. To extend or tune an existing model (e.g., adding D-M to an A-C model), use 'tune.py'.

```
python src/train.py # For new training
```

```
python src/tune.py # For expanding/tuning
```

6. Running Inference

Run the inference script for real-time testing:

```
python src/inference.py
```

Alternatively, launch the web-based translation interface:

```
streamlit run src/app.py
```